

Answer **all** questions.

1. The table below gives information about seven elements, **A-G**.

| Element  | Melting point (°C) | Boiling point (°C) | Electrical conductivity | Malleability |
|----------|--------------------|--------------------|-------------------------|--------------|
| <b>A</b> | 839                | 1484               | good                    | good         |
| <b>B</b> | −23                | 115                | poor                    |              |
| <b>C</b> | 1414               | 3265               | poor                    | poor         |
| <b>D</b> | −102               | −34                | poor                    |              |
| <b>E</b> | 10                 | 112                | poor                    | poor         |
| <b>F</b> | −188               | −42                | poor                    |              |
| <b>G</b> | 660                | 2470               | good                    | good         |

- (a) Use information from the table to answer parts (i)-(iii).

- (i) Give the **letter** of the element that has the greatest difference between its melting point and boiling point. [1]

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- (ii) Give the **letters** of the **two** elements that are gases at room temperature, 20 °C.

Give a reason for your choice. [2]

Letters ..... and .....

Reason .....

.....



- (iii) Give the **letter** of the element that is a metalloid.

Explain your choice.

[2]

Letter .....

Explanation .....

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- (b) One of the elements is aluminium. It reacts spectacularly with iron(III) oxide in the thermit reaction.

Complete and balance the equation for the reaction between aluminium and iron(III) oxide to produce aluminium oxide and iron.

[2]

